

GRADIENT-PRESERVING ADAPTIVE TASK SAMPLING FOR MULTI-TEACHER ON-POLICY DISTILLATION

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ABSTRACT

Multi-teacher on-policy distillation (MOPD) combines several specialist teachers in one student. Many training systems use the task sampling rate as the teacher weight. Adaptive sampling then changes both the compute allocation and the training objective.

We study task allocation for a fixed teacher mixture. Gradient-Preserving Adaptive Sampling (GPAS) samples tasks with large gradient signals more often and rescales every sampled gradient by the ratio between its target weight and sampling probability. The rescaling keeps the average input gradient to the optimizer equal to the gradient of the selected MOPD objective. A short analysis gives the variance-minimizing task distribution. Raw GPAS uses the gradient norm. Adam GPAS measures the same gradient after scaling by the current AdamW second-moment state.

Our evaluation connects each claim to a direct test. In a controlled estimator experiment, Adam GPAS reduces mean-squared error to $0.698\times$ the uniform value; the raw-gradient proposal reaches $7.19\times$ the uniform value under the same optimizer scale. Matched-state probes test whether AdamW scaling improves task ranking. End-to-end MOPD runs compare GPAS with uniform sampling, full-mixture updates, and gap-based allocation under matched student-token and teacher-token budgets. A checkpoint study measures how sampled-token and full-vocabulary distillation change the GPAS task scores.

1 INTRODUCTION

Large language models often use separate post-training runs for mathematics, code, instruction following, science, and other capabilities. Multi-teacher on-policy distillation (MOPD) sends each student response to a matching specialist teacher and trains one student from the resulting token-level feedback (Ma et al., 2026a).

Task allocation matters because teacher evaluation is expensive. One update design places every task in the same optimizer step. Another design selects one task for a task-homogeneous step. Task-homogeneous steps reduce memory use and teacher switching costs. Their use raises a practical question: how often should the system select each task?

Most implementations answer that question with a data mixture or a manually chosen sampling ratio. The same ratio controls the number of gradient evaluations and the average training gradient. Recent MOPD work shows that response length and learning speed can make the effective domain contribution very different from the prompt ratio (Gao et al., 2026). Increasing the sampling rate of a weak task changes its influence on the final model. Some training goals favor that change. Deployment requirements may instead specify a fixed teacher mixture.

We separate two choices. Target weights describe the MOPD objective. Sampling probabilities describe the use of gradient and teacher evaluations. When task i is sampled, a simple weight ratio converts its gradient into an estimate of the target mixture gradient. The training system can then adapt its task allocation while retaining the selected objective.

Gradient-Preserving Adaptive Sampling (GPAS) chooses the allocation from recent task gradients. The task score measures the size of the update signal, including stochastic variation inside a task. Sampling in proportion to the score minimizes the task-level gradient estimator error for the current

model. Raw GPAS uses the usual gradient norm. Adam GPAS first scales the gradient with the current AdamW second-moment state. The Adam-scaled score reflects the coordinates where the optimizer can make a larger parameter update.

The paper makes three contributions:

- We formulate task sampling as a compute-allocation decision for a fixed, response-normalized MOPD objective. The formulation separates target weights from task sampling probabilities.
- We derive GPAS with one short variance calculation and give two practical scores: raw-gradient GPAS and Adam-scaled GPAS.
- We evaluate the analysis through gradient estimator error, matched-state AdamW probes, sampled-token versus full-vocabulary comparisons, and end-to-end fixed-mixture MOPD training.

Section 3 defines the fixed-mixture objective and the task-level estimator. Section 4 presents the GPAS allocation. Section 5 tests target preservation, estimator error, and training efficiency.

2 RELATED WORK

Multi-teacher on-policy distillation. MOPD trains a shared student on student-generated responses and routes each response to a domain teacher (Ma et al., 2026a). PromptSD uses fixed per-task rollout counts and adjusts the counts for response length and task difficulty (Ma et al., 2026b). Open-MOPD makes a closely related distinction between a target token budget and the token share observed in a mixed batch (Gao et al., 2026). Its token-share correction reweights losses inside a batch. Its gap-following rule changes domain weights during training. GPAS works at the task-batch level and keeps a user-selected teacher mixture fixed.

Task allocation in multi-task RL. Multi-task methods often change task weights to improve weak tasks or balance gradient scales (Chen et al., 2018; Corrado et al., 2026). MT-GRPO learns a task distribution and uses resampling to match the accepted batch to that distribution (Ramesh et al., 2026). GPAS starts from a fixed target distribution and adapts the proposal used to estimate its gradient. The target distribution and the proposal serve separate roles.

Importance sampling for SGD. Importance sampling allocates more computation to examples with large gradient signals while preserving a chosen data objective (Alain et al., 2015; Katharopoulos & Fleuret, 2018). GPAS applies the same principle to task-homogeneous MOPD batches. The MOPD setting adds two practical questions: teacher calls have a task label, and AdamW can rescale task gradients in different ways.

Method	Fixed target	Adaptive task choice	Correction	Adam score
Standard MOPD	implicit	no	no	no
Open-MOPD share balance	yes	no	batch loss	no
MT-GRPO	learned	yes	resampling	no
Raw GPAS	yes	yes	weight ratio	no
Adam GPAS	yes	yes	weight ratio	yes

Table 1: GPAS separates a fixed target mixture from the task proposal used for gradient evaluation. The correction column describes how each method aligns the realized update with its target.

3 FIXED-MIXTURE MOPD

3.1 TARGET TEACHER MIXTURE

Consider M tasks. Task i has a prompt distribution $\mathcal{D}_i$ and a frozen teacher τ_i . The student policy is π_θ . Let $L_i(\theta)$ be the response-normalized distillation loss for task i :

$$L_i(\theta) = \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T D_{\text{KL}}(\pi_\theta(\cdot | c_t) \| \tau_i(\cdot | c_t)) \right]. \quad (1)$$

The expectation covers prompts from $\mathcal{D}_i$ and responses sampled from the student. The context c_t contains the prompt and the response prefix. Response normalization gives every response equal weight before tasks are combined. The optimizer uses the standard MOPD semi-gradient: generated responses and prefix contexts stay fixed during each parameter update.

The target weights are $\lambda_1, \dots, \lambda_M$, with $\sum_i \lambda_i = 1$. The training objective is

$$L_\lambda(\theta) = \sum_i \lambda_i L_i(\theta). \quad (2)$$

The vector λ expresses a deployment or capability priority. We choose λ independently of dataset size, response length, and task sampling frequency.

3.2 TASK SAMPLING

Let G_i be a stochastic gradient for task i , after response normalization, and let $g_i = \mathbb{E}[G_i]$. A full-mixture update uses $\sum_i \lambda_i G_i$. A task-homogeneous update first samples one task:

$$I \sim q, \quad \hat{G} = \frac{\lambda_I}{q_I} G_I. \quad (3)$$

The proposal q controls the fraction of gradient evaluations assigned to each task. Averaging Equation 3 over the sampled task gives

$$\mathbb{E}[\hat{G}] = \sum_i \lambda_i g_i. \quad (4)$$

The weight ratio in Equation 3 preserves the target mixture. Unit gradient scale gives the average gradient $\sum_i q_i g_i$ and uses the proposal as the effective teacher mixture.

Our implementation uses task-homogeneous microbatches with a fixed response token budget. Each task loss is averaged before applying λ_I/q_I . The same definition supports sampled-token and full-vocabulary distillation gradients.

4 GRADIENT-PRESERVING ADAPTIVE SAMPLING

GPAS chooses the proposal q in Equation 3. The target weights λ remain fixed.

4.1 SAMPLING SCORE

Fix the current student and optimizer state. Let v be the lagged, bias-corrected AdamW second-moment estimate before the next task is sampled. For a task gradient G_i , define the Adam-scaled gradient by coordinatewise division:

$$S_i = \frac{G_i}{\sqrt{v} + \epsilon}. \quad (5)$$

The score for task i is

$$s_i = \sqrt{\mathbb{E} \|S_i\|_2^2}. \quad (6)$$

The score includes the mean update signal and stochastic variation within the task. GPAS uses recent task gradients to maintain a moving estimate of s_i .

4.2 TASK PROPOSAL

For the corrected gradient, the Adam-scaled second moment is

$$\mathbb{E} \left\| \frac{\lambda_I}{q_I} S_I \right\|_2^2 = \sum_i \frac{\lambda_i^2 s_i^2}{q_i}. \quad (7)$$

Equation 4 fixes the average gradient. Reducing Equation 7 also reduces the mean-squared error of the gradient estimate in the same Adam-scaled coordinates. The minimum uses

$$q_i^* = \frac{\lambda_i s_i}{\sum_j \lambda_j s_j}. \quad (8)$$

Appendix A.2 gives a short proof.

Adam GPAS estimates the score in Equation 6. Raw GPAS replaces S_i with G_i . Both methods apply λ_I/q_I before the AdamW moment update. A probability floor keeps every teacher active and limits large weight ratios.

The gradient guarantee applies at the input to AdamW. AdamW updates its second-moment state with squared, importance-weighted gradients, so different proposals produce different optimizer paths. Section 5.4 tests whether the lagged Adam-scaled score predicts the next task-dependent parameter step.

4.3 SAMPLED-TOKEN AND FULL-VOCABULARY GRADIENTS

GPAS uses the gradient produced by the selected distillation estimator. Sampled-token OPD estimates each token loss from the rollout token. Full-vocabulary OPD sums the loss over the vocabulary at the same response prefix. The vocabulary sum removes token-sampling noise and requires more teacher output.

The optimal proposal keeps the form in Equation 8 for both estimators. The measured scores can differ. Sampled-token scores include token-sampling variation, while full-vocabulary scores place more emphasis on prompt and response variation. Section 5.5 measures the change in task ranking, estimator error, and proposal entropy on matched response prefixes.

5 EXPERIMENTS

The experiments test four claims. The weight ratio preserves the selected teacher mixture. The GPAS proposal reduces task-level estimator error. The Adam-scaled score can predict task-dependent AdamW updates better than the raw gradient norm. Sampled-token and full-vocabulary distillation can produce different task scores.

5.1 SETUP

We use Qwen3-1.7B as the shared initialization. Four same-origin teachers are trained from that checkpoint with single-task GRPO on mathematics, code, instruction following, and science question answering. The MOPD student starts from the shared initialization. Every method uses the same frozen teachers, prompt pools, response limits, AdamW configuration, and learning-rate schedule.

The primary MOPD loss uses the sampled-token reverse-KL estimator. Each task microbatch contains about 65,536 generated response tokens. The loss first averages tokens within a response and responses within a task microbatch. Every end-to-end run uses 600 optimizer updates. The target mixture is uniform, with $\lambda_i = 1/4$.

We use seeds 41, 42, and 43 for the main training methods. Teacher loss is measured every 25 updates on fixed held-out prompts. Capability evaluation runs at updates 0, 150, 300, 450, and 600. We report generated response tokens, teacher-scored tokens, wall-clock time, and GPU time for every run.

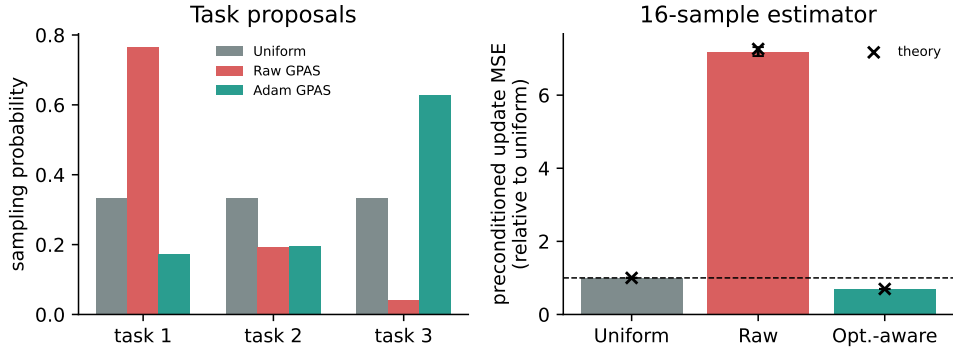


Figure 1: Controlled test of task allocation. Bars show 20,000 Monte Carlo trials, and crosses show the value from Equation 7. Adam GPAS uses the task proposal predicted by the current optimizer scale and has the lowest estimator error.

5.2 CONTROLLED ESTIMATOR TEST

We first isolate the sampling result from language-model training. The controlled problem has three equally weighted tasks and 96 gradient coordinates. Each task has most of its gradient variation in a different coordinate block. A fixed AdamW scale assigns different update sizes to the three blocks.

Uniform sampling, Raw GPAS, and Adam GPAS estimate the same target gradient from 16 sampled tasks. We compare the formula in Equation 7 with 20,000 Monte Carlo trials. Figure 1 shows close agreement between the formula and simulation. Raw GPAS assigns too much probability to a large raw gradient that AdamW strongly suppresses. Its estimator error is $7.19\times$ the uniform error. Adam GPAS follows the Adam-scaled task scores and reduces the error to $0.698\times$ the uniform value.

5.3 TARGET PRESERVATION AND ESTIMATOR ERROR

At updates 0, 150, 300, 450, and 600 of the seed-42 uniform run, we collect 32 independent fixed-token gradients per task. A large stratified pool estimates the full target gradient. The same gradient pool supports Monte Carlo evaluation of each task proposal.

We compare corrected and uncorrected estimates from the same proposal. The primary target-preservation measures are relative gradient bias and cosine similarity with the full target gradient. The primary allocation measure is the Adam-scaled mean-squared error:

$$\mathbb{E} \left\| \frac{\hat{G} - g_{\lambda}}{\sqrt{v} + \epsilon} \right\|_2^2. \quad (9)$$

The comparison uses uniform sampling, gap-based sampling, Raw GPAS, Adam GPAS, and a retrospective oracle computed from the full gradient pool. The oracle measures the error caused by online score estimation.

The corrected and uncorrected gap proposals share the same task sequence and score history. Their difference isolates the weight ratio in Equation 3. We also report the cumulative gradient weight assigned to every task. The cumulative weights make objective drift visible without relying on a final benchmark score.

5.4 DOES ADAMW SCALING IMPROVE TASK RANKING?

We copy the model and AdamW state before every one-step probe. Each task provides 16 independent fixed-token gradients at each evaluation checkpoint. For every gradient, we record its raw norm, Adam-scaled norm, and the exact parameter displacement produced by one AdamW step. A zero-gradient clone removes displacement from stored momentum and weight decay.

We measure Spearman correlation and pairwise task-ranking accuracy between each score and the task-dependent parameter displacement. Held-out teacher loss reduction provides a secondary mea-

sure that also depends on update direction. The same checkpoints compare the proposal predicted by each score with the retrospective oracle from Section 5.3.

The matched-state design tests the Adam approximation used by GPAS. The end-to-end comparison between Raw GPAS and Adam GPAS tests whether better one-step ranking improves training efficiency.

5.5 SAMPLED-TOKEN VERSUS FULL-VOCABULARY OPD

We compare the two distillation estimators on the same prompts, response prefixes, student checkpoint, and AdamW state. The full-vocabulary gradient uses the complete student and teacher distributions at each selected prefix. The sampled-token gradient draws tokens from the same student distribution. Repeated token draws estimate the token-sampling part of the gradient variation.

For each task and checkpoint, we report four quantities: the raw GPAS score, the Adam GPAS score, the resulting task probability, and estimator error. Rank correlation between sampled-token and full-vocabulary scores shows whether both estimators prioritize the same teachers. Proposal entropy shows whether removing token-sampling noise makes the task allocation more uniform. Teacher-logit memory, teacher-token throughput, and wall-clock time measure the cost of the full-vocabulary signal.

5.6 END-TO-END MOPD EFFICIENCY

All methods optimize the same uniform teacher mixture unless the method name states an uncorrected ablation:

1. **Uniform:** sample every task with probability $1/4$.
2. **Full mixture:** use one quarter of each update’s token budget for every task and average the four task gradients.
3. **Corrected gap:** sample from a teacher-gap score and apply λ_I/q_I .
4. **Uncorrected gap:** use the same proposal with unit gradient scale.
5. **Raw GPAS:** use the raw-gradient score and apply λ_I/q_I .
6. **Adam GPAS:** use the Adam-scaled score and apply λ_I/q_I .

Adaptive methods start with 16 round-robin updates, giving four observations per task. Each method uses an exponential moving average with decay 0.95 and a probability floor of 0.05. The importance ratio is applied before the AdamW first- and second-moment updates.

We report weighted teacher-loss area under the training curve, tokens needed to reach fixed loss thresholds, and wall-clock time. Per-task teacher losses at matched weighted loss measure task balance. Final evaluations use MATH-500, LiveCodeBench, IFBench, and GPQA-Diamond. The main GPAS claim requires lower estimator error and faster fixed-mixture training under matched student-token and teacher-token budgets.

6 PRACTICAL USE AND SCOPE

Choose the target mixture first. Set λ from deployment priorities or a capability target. Average each response loss over its generated tokens before applying the task weight. The target mixture then has a clear task-level meaning.

Use full-mixture updates when every teacher fits in one step. A full-mixture update removes task-selection noise. GPAS is useful when an update uses one teacher, when teacher switching is expensive, or when the number of teachers makes full accumulation slow.

Warm up the task scores. Start with a short round-robin period and collect several gradients from every task. Update each score after the sampled task finishes. A probability floor keeps rare-task scores fresh and limits the importance ratio.

Apply the weight ratio before AdamW. For sampled task I , multiply its response-normalized gradient by λ_I/q_I before updating the AdamW moments. Log the target weight, proposal, correction factor, score, and score age at every step.

Select the score with a one-step test. Raw GPAS has the simplest implementation. Adam GPAS is useful when the Adam-scaled norm ranks task-dependent parameter steps more accurately in the matched-state probe from Section 5.4. The one-step test provides a low-cost choice before an end-to-end run.

Scope. The analysis covers fixed-token, task-homogeneous updates and a fixed teacher mixture. The gradient result is conditional on the current model and optimizer state. AdamW state, rollouts, and task gradients continue to change during training, so the end-to-end evaluation measures the final efficiency claim. The sampled-token and full-vocabulary estimators use the same GPAS rule and maintain separate task scores.

7 CONCLUSION

MOPD systems make two separate choices: the teacher mixture defines the training objective, and the task proposal allocates teacher and gradient evaluations. GPAS connects the two choices with a simple importance weight. The resulting gradient estimate preserves the selected mixture.

GPAS allocates more updates to tasks with large optimizer-scaled gradient signals. The allocation minimizes task-level estimator error for the current model and AdamW state. Raw GPAS provides a simple default. Adam GPAS uses the optimizer state when AdamW changes the relative scale of task updates.

The experimental design tests the method at three levels: gradient estimator error, matched-state optimizer behavior, and end-to-end MOPD efficiency. The sampled-token and full-vocabulary study also shows how the distillation estimator changes the task scores used for allocation.

AI USE STATEMENT

We used generative AI tools for language editing and manuscript organization. The authors reviewed every AI-assisted revision and take responsibility for the final text, claims, proofs, code, and experimental results.

ETHICS STATEMENT

The study evaluates optimization methods for existing language models with established reasoning and instruction-following datasets and automated verifiers. Model and dataset biases can affect the reported capability scores. We record data provenance, invalid outputs, and verifier failures to support careful interpretation of the results.

REPRODUCIBILITY STATEMENT

Section 5 specifies the model family, task setup, update budget, evaluation points, and evaluation protocol. The appendix gives the complete derivations. The supplementary experiment record includes frozen data manifests, model and tokenizer revisions, random seeds, optimizer settings, task-sampling logs, saved parameter and optimizer states, and per-prompt evaluation outputs.

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A ADDITIONAL ANALYSIS AND EXPERIMENTAL DETAILS

A.1 TARGET-GRADIENT CALCULATION

For task $I \sim q$, the corrected gradient from Equation 3 has mean

$$\mathbb{E}[\widehat{G}] = \sum_i q_i \frac{\lambda_i}{q_i} \mathbb{E}[G_i] = \sum_i \lambda_i g_i. \quad (10)$$

The proposal cancels from the average gradient. A proposal may depend on all completed training steps. The next task is sampled after the proposal has been set.

A.2 OPTIMAL GPAS PROPOSAL

Let $s_i^2 = \mathbb{E}\|S_i\|_2^2$. The Adam-scaled second moment of the corrected gradient is

$$\sum_i \frac{\lambda_i^2 s_i^2}{q_i}. \quad (11)$$

Cauchy–Schwarz gives

$$\left(\sum_i \lambda_i s_i \right)^2 \leq \left(\sum_i q_i \right) \left(\sum_i \frac{\lambda_i^2 s_i^2}{q_i} \right). \quad (12)$$

The first factor on the right is one. Equality requires q_i to be proportional to $\lambda_i s_i$. Normalizing the probabilities gives Equation 8.

The implementation adds a probability floor. Tasks below the floor receive the floor value, and the remaining probability is distributed in proportion to $\lambda_i s_i$. Four-task water filling solves the allocation with a few scalar operations.

A.3 ADAMW SCORE

Adam GPAS uses the second-moment state stored before task sampling. Dividing a task gradient by $\sqrt{v} + \epsilon$ estimates the coordinate scale used by the next AdamW update. Stored momentum and weight decay contribute a common movement in the one-step probe. A zero-gradient clone removes that common movement.

The target-gradient calculation in Equation 10 covers the gradient sent into AdamW. The sampled gradient also updates v , so each proposal creates its own AdamW state path. The end-to-end experiment measures the effect of that state path.

A.4 SAMPLED-TOKEN AND FULL-VOCABULARY SCORES

For one fixed response prefix, let X_i be the sampled-token semi-gradient and let F_i be its average over the student’s token distribution. The vector F_i is the corresponding full-vocabulary semi-gradient for the same prefix. With the AdamW denominator held fixed,

$$\mathbb{E} \left\| \frac{X_i}{\sqrt{v} + \epsilon} \right\|_2^2 = \mathbb{E} \left\| \frac{F_i}{\sqrt{v} + \epsilon} \right\|_2^2 + \mathbb{E} \left\| \frac{X_i - F_i}{\sqrt{v} + \epsilon} \right\|_2^2. \quad (13)$$

The last term measures token-sampling variation. Full-vocabulary OPD removes that term at each prefix. Prompt and rollout variation remain in both estimators. The checkpoint experiment estimates every term with matched prefixes and repeated token draws.

A.5 CONTROLLED ESTIMATOR PROBLEM

The controlled problem uses three tasks, 96 coordinates, and a uniform target mixture. Coordinates are split into three blocks of size 32. Task-gradient standard deviations in the dominant blocks are 2.0, 0.5, and 0.1. The fixed AdamW scales for the same blocks are 0.005, 0.05, and 3.0. The raw-gradient score favors the first task, while the Adam-scaled score favors the third task.

Every trial averages 16 independently sampled, importance-corrected task gradients. We use 20,000 trials and report squared error after applying the fixed AdamW scale. The script `experiments/controlled_optimizer_sampling.py` reproduces Figure 1.

A.6 END-TO-END IMPLEMENTATION DETAILS

The adaptive methods use four round-robin observations per task before score updates begin. For a sampled task, the implementation performs the following operations:

1. average the loss over valid response tokens and responses;
2. compute the task gradient and its GPAS score;
3. multiply the gradient by λ_I/q_I ;
4. send the corrected gradient directly to the AdamW moment update.

The primary target-preservation runs disable gradient clipping, so Equation 10 applies to the gradient received by AdamW. The probability floor limits large correction factors.

Each run logs the sampled task, proposal, target weight, importance ratio, score, score age, response-token count, teacher-token count, gradient norm, and Adam-scaled norm. Checkpoints also save the proposal moving averages and task-sampler random state.